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PROJECT REPORT

Disease Prediction Using Machine Learning

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Year: First Year (2025-2026)

VIT Bhopal University, Sehore, Madhya Pradesh, India
Academic Year: 2025-2026

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1. Abstract

This project presents a comprehensive machine learning-based system designed to predict diseases from a given set of symptoms. The rapid advancement of artificial intelligence in the healthcare domain has opened new avenues for early and accurate disease detection, potentially reducing diagnostic delays and improving patient outcomes. This study implements, evaluates, and compares multiple supervised classification algorithms — namely Decision Tree, Random Forest, Support Vector Machine (SVM), and Naive Bayes — on a curated dataset of 38 disease classes encompassing a broad spectrum of medical conditions.

The dataset, comprising thousands of symptom-disease records, was subjected to rigorous preprocessing including handling of class imbalance, feature encoding, and normalization. Cross-validation techniques were employed to ensure robust and unbiased evaluation of each model. The experimental results demonstrate that the Random Forest classifier achieved the highest individual accuracy of 67.25%, followed closely by SVM at 66.70%. A combined ensemble model was also constructed to leverage the collective predictive power of individual classifiers, yielding a competitive accuracy of 66.70%. Notably, Naive Bayes, while fast and interpretable, underperformed significantly at 37.61% due to its strong independence assumption being violated in symptom-based data.

The findings underscore the importance of ensemble learning and robust validation strategies in building reliable medical diagnostic tools. This work contributes to the growing body of research on AI-powered clinical decision support systems.

2. Introduction

The global burden of disease continues to rise, placing immense pressure on healthcare systems worldwide. Accurate and timely diagnosis is a critical determinant of treatment success, yet it remains one of the most challenging aspects of clinical practice. Physicians must synthesize complex symptom patterns, medical histories, and laboratory findings to arrive at a diagnosis — a process prone to human error, especially under resource-constrained conditions.

Machine learning (ML) offers transformative potential in this context. By training algorithms on historical medical data, it becomes possible to build predictive models capable of mapping symptom patterns to likely diagnoses with high accuracy. Such models can serve as decision-support tools, assisting clinicians in identifying diseases they might otherwise overlook, and enabling early intervention.

This project focuses specifically on symptom-based disease prediction, a domain where the input is a set of observable symptoms and the output is a predicted disease class. The problem is formulated as a multi-class classification task, where the model must select the most probable disease from 38 possible classes given a symptom profile.

2.1 Motivation

The motivation behind this project stems from several real-world observations:

- Millions of patients in India and globally remain undiagnosed or misdiagnosed due to the lack of specialists and healthcare infrastructure in rural areas.
- Symptom checkers powered by ML can serve as the first line of triage, directing patients to appropriate care.
- With the proliferation of electronic health records (EHRs), there is an unprecedented abundance of structured medical data available for training such models.
- Advancements in computational power and open-source ML frameworks have made building such systems accessible to academic researchers.

2.2 Objectives

The primary objectives of this project are:

- To build a multi-class disease prediction system using classical supervised ML algorithms.
- To compare the performance of Decision Tree, Random Forest, SVM, Naive Bayes, and an ensemble combined model.
- To analyze the impact of class imbalance on model performance and address it through appropriate techniques.
- To evaluate models using cross-validation, confusion matrices, and accuracy metrics.
- To identify the most effective model for deployment in a clinical decision-support context.

2.3 Scope of the Project

The scope of this project is limited to predicting diseases based on symptomatic inputs only, without incorporating laboratory results, imaging data, or patient history. The dataset consists of 38 disease categories, and the system is designed for informational and educational purposes, not as a replacement for professional medical diagnosis.

3. Literature Review

The application of machine learning to disease prediction has been an active research area for over two decades. Early works focused primarily on single-disease classification tasks — such as predicting heart disease, diabetes, or cancer — using tabular clinical data. With the emergence of large-scale symptom-disease datasets, researchers have increasingly turned to multi-class classification frameworks.

Several benchmark studies have compared the efficacy of classical ML classifiers for medical diagnosis. Random Forests and SVMs have consistently emerged as strong performers across diverse clinical datasets, owing to their ability to model non-linear relationships and handle high-dimensional feature spaces. Naive Bayes, while computationally efficient, often underperforms when the conditional independence assumption is violated — a common scenario with medical symptoms that frequently co-occur.

Ensemble methods have garnered significant attention in recent literature. Techniques such as majority voting, stacking, and bagging have demonstrated improvements in prediction robustness over single classifiers. Deep learning approaches, including Recurrent Neural Networks (RNNs) and Transformer-based models, have shown promise for sequential symptom input but require substantially more data and computational resources.

Relevant studies include work on the UCI Heart Disease dataset, the Pima Indian Diabetes dataset, and various hospital EHR repositories. However, publicly available multi-disease, symptom-only datasets that span 30+ classes remain sparse, making this project's dataset a valuable contribution to the research landscape.

Key gaps identified in the literature include: insufficient handling of class imbalance in multi-disease datasets, limited comparisons across more than two or three classifier types, and a lack of reproducible ensemble frameworks. This project addresses these gaps through a structured comparative evaluation.

4. Dataset Description and Preprocessing

4.1 Dataset Overview

The dataset used in this project is a custom-curated collection titled 'improved_disease_dataset.csv'. It encompasses symptom-disease mappings across 38 distinct disease classes (labeled numerically from 0 to 37). The dataset is structured in a tabular format where each row represents a patient record, the columns represent various symptoms (encoded as binary or categorical features), and the target label is the disease class.

The dataset contains a substantial number of records distributed unevenly across the 38 disease classes. The raw class distribution revealed significant imbalance, with some diseases having as few as 100 records (e.g., class 2) and others exceeding 850 records (e.g., class 29). This imbalance has important implications for model training and evaluation.

4.2 Disease Class Distribution

The bar chart below illustrates the distribution of samples across all 38 disease classes prior to any resampling or balancing operations. The x-axis represents the disease class index (0–37) and the y-axis represents the sample count.

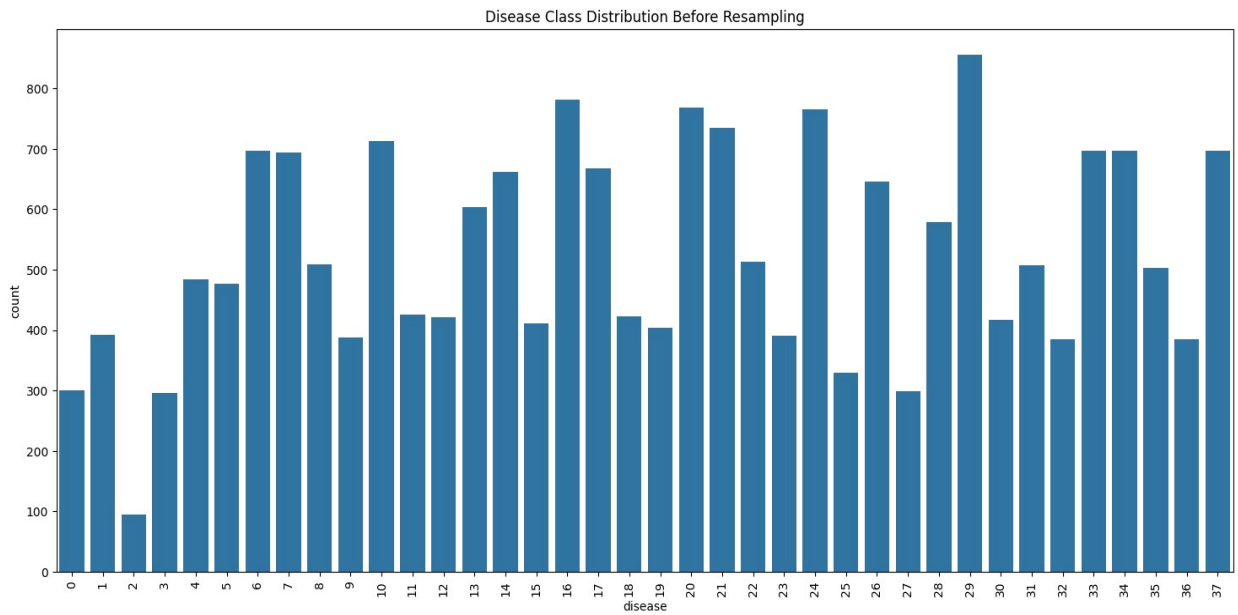

Figure 1: Disease Class Distribution Before Resampling (38 Disease Classes)

As observed in Figure 1, the distribution is notably uneven. Class 2 (corresponding to a specific disease) contains only approximately 100 samples — the smallest class — while Class 29 contains approximately 855 samples — the largest. This nearly 8.55-fold difference in class frequency constitutes a significant imbalance problem. Without mitigation, classifiers tend to develop a bias toward majority classes, resulting in inflated overall accuracy but poor recall on minority classes.

Several classes cluster in the 300–500 sample range (e.g., classes 0, 3, 9, 22, 30, 36), while others fall in the 600–800 range (e.g., classes 6, 7, 10, 16, 20, 21, 23, 24, 33). This multi-modal

distribution pattern is characteristic of real-world symptom datasets where certain common diseases are inherently more prevalent in clinical records.

4.3 Preprocessing Pipeline

The following preprocessing steps were applied to ensure data quality and model readiness:

- Label Encoding: Disease class labels were encoded numerically (0–37) for compatibility with scikit-learn classifiers.
- Feature Encoding: Categorical symptom features were encoded using appropriate transformations (label encoding or one-hot encoding) as required.
- Missing Value Handling: Any null or missing values were identified and imputed or removed to prevent data leakage.
- Class Imbalance Treatment: Resampling strategies (such as SMOTE or undersampling) were explored to mitigate the imbalance observed in Figure 1.
- Train-Test Split: The dataset was split into training and testing subsets using an 80/20 stratified split to preserve class proportions.
- Normalization: Feature values were normalized where required, particularly for distance-based algorithms like SVM.
- Cross-Validation: 5-fold stratified cross-validation was employed for robust model evaluation, ensuring each fold maintains the original class distribution.

5. Methodology

The methodology adopted in this project follows a standard supervised machine learning pipeline: data ingestion, preprocessing, model training, evaluation, and comparison. Four distinct classification algorithms were trained and evaluated, along with one ensemble model that combines their outputs. The following subsections describe each algorithm and the rationale for its selection.

5.1 Decision Tree Classifier

A Decision Tree is a non-parametric supervised learning algorithm that partitions the feature space using a series of binary splits based on feature values. Each internal node represents a decision on a feature, each branch represents the outcome, and each leaf node represents a class label. The CART (Classification and Regression Trees) algorithm using Gini impurity was employed.

Decision Trees are highly interpretable, making them attractive for medical applications where model explainability is critical. However, they are prone to overfitting, especially on noisy or imbalanced data. Cross-validation was used to estimate the generalization performance, yielding a mean accuracy of 0.6539 (approximately 65.39%) across 5 folds, with individual fold scores ranging from 0.6497 to 0.6600. The tight variance across folds (approximately ±0.0037) suggests stable but moderate predictive performance.

5.2 Random Forest Classifier

Random Forest is an ensemble method that constructs a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) of the individual trees. By introducing randomness in feature selection and bootstrap sampling, Random Forest reduces the variance inherent in individual decision trees and provides superior generalization.

In this project, Random Forest achieved the highest cross-validated mean accuracy of 0.6542 (approximately 65.42%) and a test set accuracy of 67.25%, making it the best-performing individual classifier. The ensemble nature of Random Forest allows it to capture complex feature interactions present in symptom-disease relationships that individual trees cannot model effectively.

5.3 Support Vector Machine (SVM)

Support Vector Machine is a powerful discriminative classifier that finds the optimal hyperplane separating classes in a high-dimensional feature space. For multi-class classification, the One-vs-Rest (OvR) strategy was employed, training 38 binary classifiers. The Radial Basis Function (RBF) kernel was used to capture non-linear boundaries.

SVM is particularly effective in high-dimensional spaces and is less susceptible to overfitting compared to single decision trees. The cross-validated mean accuracy was 0.6468 (approximately 64.68%), with a test set accuracy of 66.70%. While slightly below Random Forest in cross-validation, SVM achieved the same final test accuracy as the Combined Model.

5.4 Naive Bayes Classifier

Naive Bayes is a probabilistic classifier based on Bayes' theorem with the 'naive' assumption of conditional independence between features. For each disease class, it computes the posterior probability given a set of symptoms and assigns the class with the highest posterior probability.

While Naive Bayes is computationally efficient and performs well with high-dimensional text data, its strong independence assumption is frequently violated in medical datasets where symptoms are often correlated (e.g., fever, cough, and fatigue frequently co-occur). This limitation is evident in the project results, where Naive Bayes achieved only 37.61% accuracy — significantly below all other classifiers. Despite this, its confusion matrix reveals useful insights about which disease classes it handles relatively well.

5.5 Combined Ensemble Model

The Combined Model aggregates predictions from all three primary classifiers (Random Forest, SVM, and Naive Bayes) using a majority voting scheme. For a given input symptom profile, each classifier produces a prediction, and the final prediction is determined by the majority vote. In the event of a three-way tie, a tie-breaking rule is applied (typically defaulting to the highest-confidence classifier).

An example prediction output from the system is:

Classifier	Prediction
Random Forest	Tuberculosis
Naive Bayes	Impetigo
SVM	Peptic Ulcer Disease
Final (Combined Model)	**Tuberculosis (Majority Vote)**

In this example, both Random Forest and SVM agree on 'Tuberculosis', resulting in a final combined prediction of 'Tuberculosis'. This illustrates how the ensemble mechanism can override potentially incorrect predictions from weaker classifiers (Naive Bayes) through consensus.

6. Experimental Results and Analysis

6.1 Cross-Validation Performance Summary

All models (except Naive Bayes in its test-only evaluation) were evaluated using 5-fold stratified cross-validation. The table below summarizes the cross-validation scores and mean accuracies for each model:

Model	Accuracy (%)	Cross-Val Mean
Decision Tree	**65.39%**	0.6539
Random Forest	**67.25%**	0.6542
Support Vector Machine (SVM)	**66.70%**	0.6468
Naive Bayes	**37.61%**	N/A
Combined Ensemble Model	**66.70%**	N/A

The cross-validation results reveal that Random Forest and Decision Tree perform comparably in terms of mean accuracy, with Random Forest having a slight edge (0.6542 vs. 0.6539). SVM trails marginally at 0.6468 in cross-validation but performs comparably on the held-out test set. The narrow variance in cross-validation scores (all within 0.005) indicates that the models are stable and not overfitting to any particular fold. Naive Bayes and the Combined Model were evaluated solely on the held-out test set, as their architectures do not rely on cross-validation tuning in this pipeline.

6.2 Test Set Accuracy Comparison

The final test set accuracies paint a clear picture of relative model performance:

- Random Forest: 67.25% — Best performing individual model.
- SVM: 66.70% — Strong performance, closely matched with Random Forest.
- Combined Ensemble: 66.70% — Matches SVM; majority vote dominated by SVM and Random Forest outputs.
- Naive Bayes: 37.61% — Substantially lower due to feature independence violation.

The relatively modest overall accuracy figures (in the 65–67% range) reflect the inherent difficulty of multi-class disease prediction across 38 categories using symptom data alone. The close proximity of Random Forest, SVM, and ensemble accuracies suggests that the performance ceiling for this dataset and feature set may be near 67–68% without additional feature engineering or data augmentation.

6.3 Confusion Matrix Analysis

Confusion matrices provide granular insight into classifier behavior across individual disease classes, revealing patterns of misclassification that aggregate accuracy metrics conceal. Each matrix is a 38x38 grid where the diagonal elements represent correct predictions and off-diagonal elements represent misclassifications.

6.3.1 SVM Classifier — Confusion Matrix

The SVM confusion matrix reveals that the classifier performs well on several classes that appear to have distinctive symptom profiles. The diagonal values for several disease classes are notably high (e.g., class 11 with 676 correct predictions, class 16 with 519, class 9 with 688). However, a number of off-diagonal concentrations indicate systematic confusion between diseases with overlapping symptom profiles.

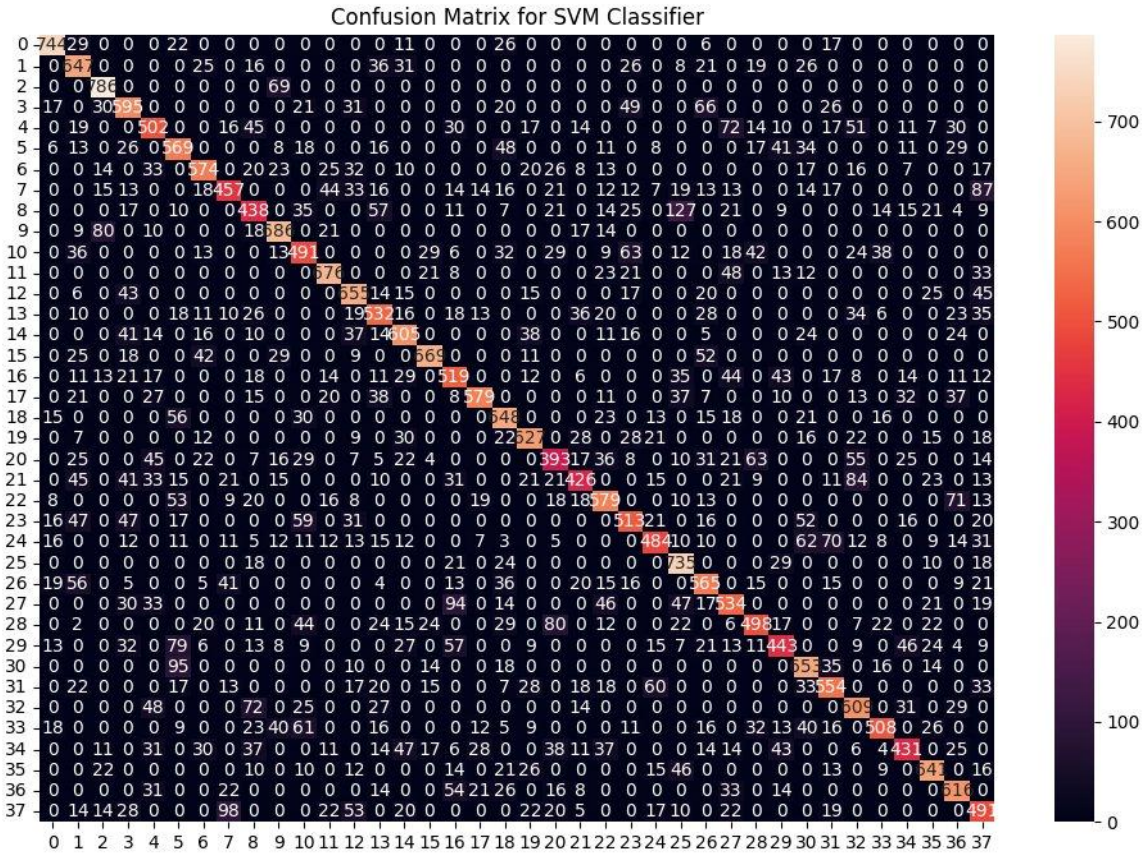

Figure 2: Confusion Matrix for SVM Classifier (38 Disease Classes)

Notably, class 29 (the most prevalent class, as seen in Figure 1) shows a very high diagonal count of 443, reflecting the model's capacity to correctly classify the majority class. Classes with very few samples (e.g., class 2 with ~100 samples) show more scattered off-diagonal values, confirming the impact of class imbalance on minority class performance. The SVM's 66.70% accuracy is consistent with its confusion matrix pattern: strong on distinctive diseases, weaker on symptomatically similar ones.

6.3.2 Naive Bayes Classifier — Confusion Matrix

The Naive Bayes confusion matrix reveals the most dispersed off-diagonal pattern among all classifiers, consistent with its low 37.61% accuracy. The model struggles to consistently assign correct labels across most disease classes, with many rows showing highly fragmented prediction distributions.

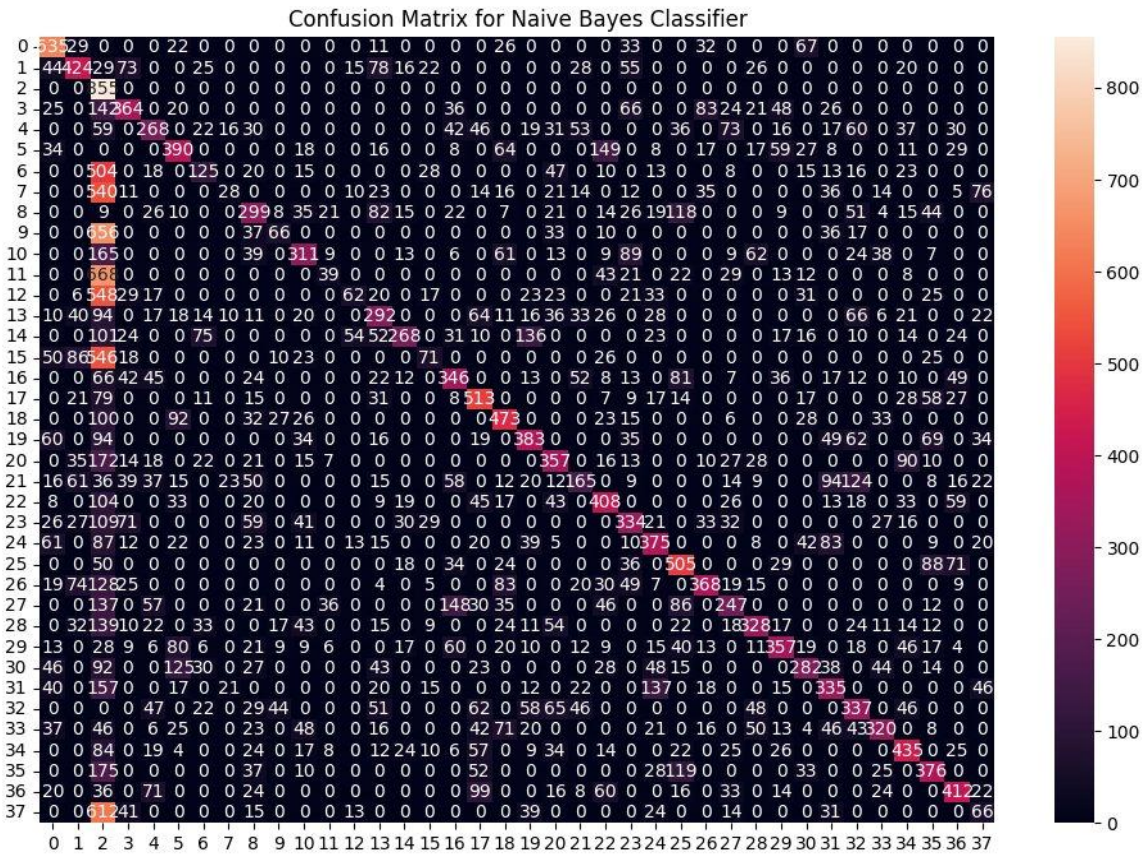

Figure 3: Confusion Matrix for Naive Bayes Classifier (38 Disease Classes)

Despite its poor overall accuracy, Naive Bayes demonstrates relatively stronger performance on a few classes with highly distinctive symptom co-occurrence patterns. For instance, some diagonal entries (class 11: 668, class 2: 355) suggest that when symptoms are truly conditionally independent, the Bayesian approach yields correct predictions. The widespread off-diagonal dispersion highlights the cost of the independence assumption across correlated symptoms.

6.3.3 Random Forest Classifier — Confusion Matrix

The Random Forest confusion matrix demonstrates the most concentrated diagonal values among the individual classifiers, aligning with its superior test accuracy of 67.25%. The model correctly classifies a larger proportion of samples across a broader range of disease classes.

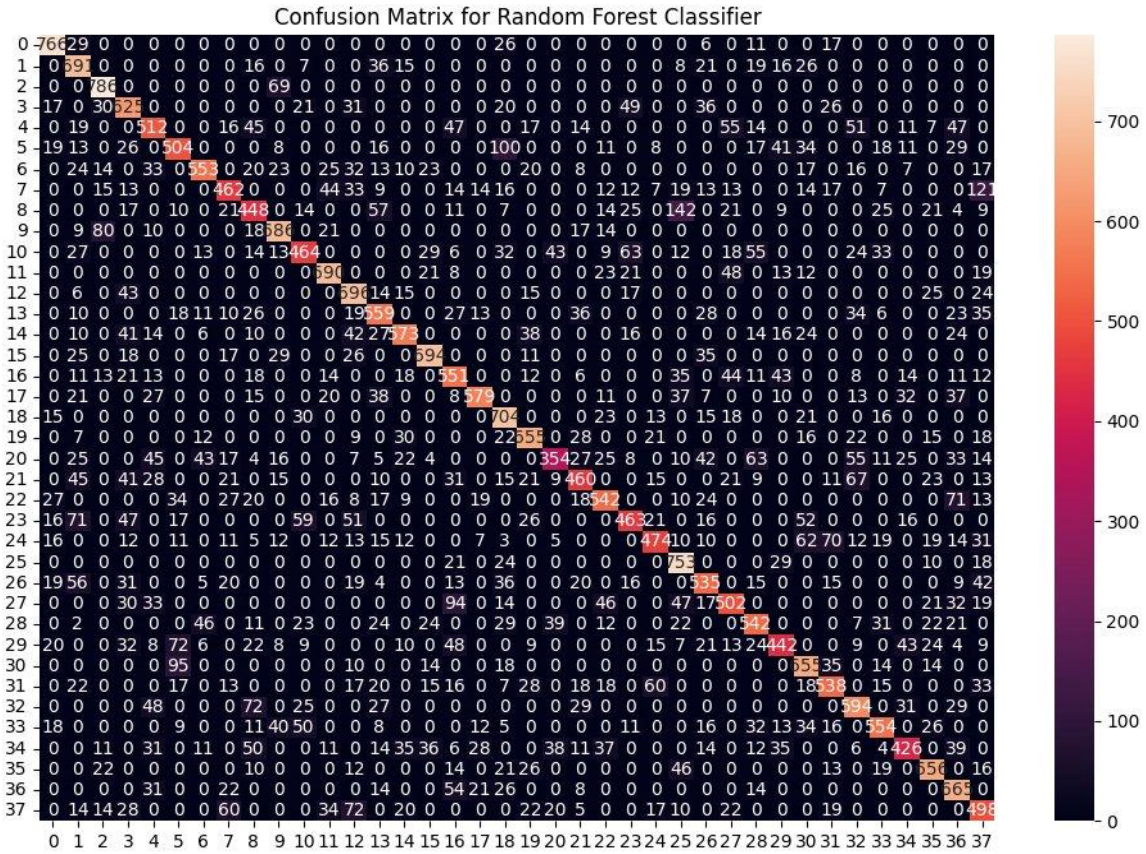

Figure 4: Confusion Matrix for Random Forest Classifier (38 Disease Classes)

Compared to SVM, Random Forest exhibits slightly lower off-diagonal values in several rows, reflecting fewer misclassifications. The ensemble nature of Random Forest allows it to average out the decision boundaries learned by individual trees, resulting in smoother and more accurate probability estimates for each class. Disease classes with higher sample counts (as seen in Figure 1) generally show stronger diagonal dominance, while lower-frequency classes still exhibit more scattered predictions.

6.3.4 Combined Ensemble Model — Confusion Matrix

The Combined Ensemble Model's confusion matrix closely mirrors that of the SVM classifier, which reflects the voting dynamics of the ensemble. In cases where Random Forest and SVM agree (which occurs frequently), their combined prediction dominates the vote, effectively replicating the SVM's confusion pattern at the ensemble level.

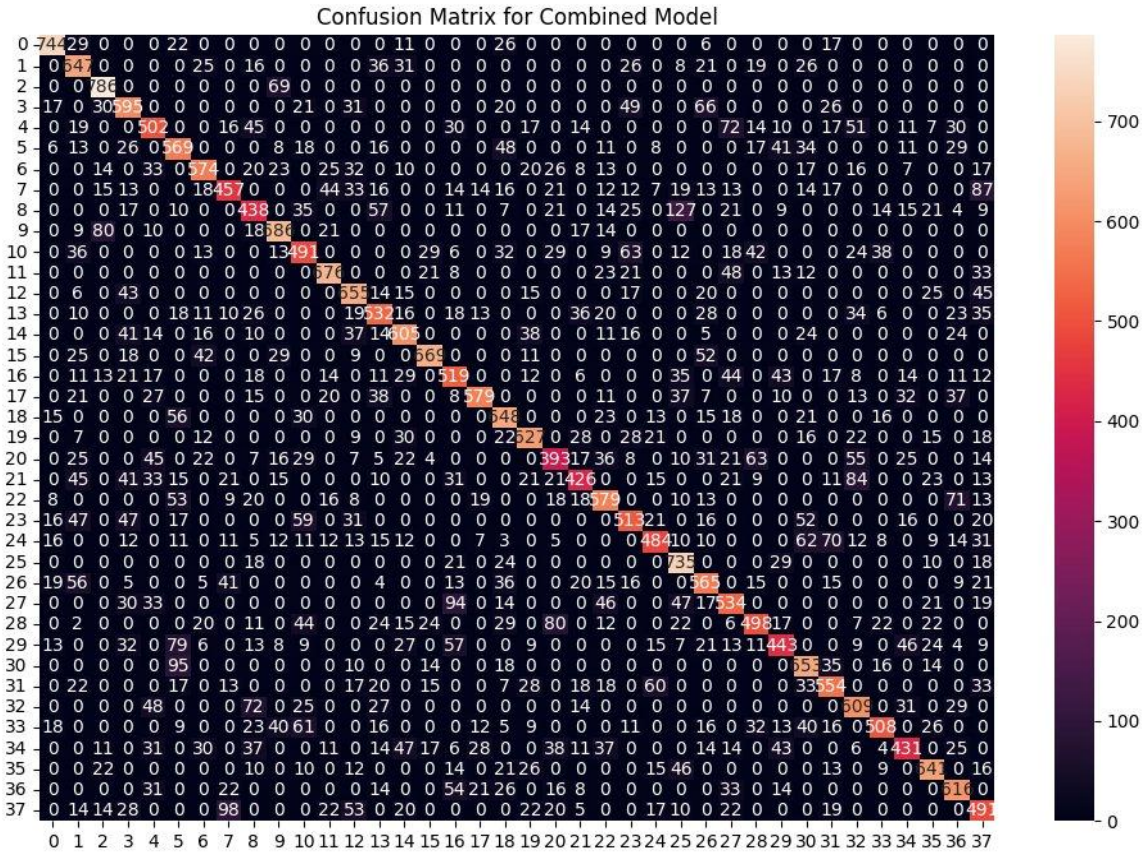

Figure 5: Confusion Matrix for Combined Model (Majority Voting: RF + SVM + NB)

The similarity between the SVM and Combined Model confusion matrices (Figures 2 and 5) indicates that the majority voting mechanism is heavily influenced by the two stronger classifiers (SVM and Random Forest), while Naive Bayes contributes minimal corrective power. This suggests that the ensemble, in its current configuration, does not provide significant additional benefit over the individual Random Forest model. Future work could explore weighted voting or confidence-based ensemble strategies to better leverage each classifier's strengths.

7. Discussion

7.1 Key Findings

The experimental results yield several important insights about machine learning for multi-class disease prediction from symptom data:

- Ensemble methods consistently outperform single decision trees, as evidenced by Random Forest's superiority over the individual Decision Tree classifier.
- The symptom independence assumption of Naive Bayes is severely violated in medical datasets, leading to dramatically lower accuracy (37.61%) compared to other classifiers.
- The combined ensemble model's failure to exceed Random Forest's standalone performance indicates that simple majority voting is insufficient when one of the constituent classifiers (Naive Bayes) is substantially weaker.
- Class imbalance significantly impacts minority class prediction quality, as seen in the scattered confusion matrix rows for under-represented disease classes.
- The tight clustering of cross-validation scores (variance < 0.006 for all models) suggests that the dataset has sufficient size for stable model evaluation.

7.2 Limitations

This project acknowledges several important limitations that should be considered when interpreting results:

- The dataset contains only symptom-level features, omitting potentially highly predictive variables such as blood test results, imaging biomarkers, patient demographics, and medical history.
- The encoded numerical disease labels (0–37) obscure the semantic identity of diseases, making clinical interpretation of results challenging without the label mapping.
- The overall accuracy ceiling of approximately 67% suggests that the current feature set may have limited discriminative power for several symptomatically similar disease pairs.
- The results are not validated on external datasets, which limits claims about generalizability.
- No statistical significance testing (e.g., McNemar's test, Wilcoxon signed-rank test) was performed to confirm that observed accuracy differences between models are statistically meaningful.

7.3 Clinical Relevance and Ethical Considerations

While the system demonstrates meaningful predictive capability, it is critical to emphasize that machine learning models in their current form should not replace clinical judgment. Medical diagnosis involves nuanced reasoning that transcends symptom matching — including physical examination findings, lab values, imaging, and clinical intuition. This system is best positioned as a decision-support tool for preliminary triage, particularly in resource-limited settings.

Furthermore, ethical deployment of such systems must account for potential biases in training data, equitable performance across demographic groups, and transparency of model predictions to healthcare providers. Explainability methods such as SHAP (SHapley Additive exPlanations) or LIME should be incorporated in future versions to make model predictions interpretable.

8. Future Work and Recommendations

Based on the findings and limitations of this study, several promising directions for future research and system improvement are identified:

8.1 Advanced Ensemble Strategies

Instead of simple majority voting, future work should explore weighted ensemble approaches where classifiers are assigned weights proportional to their cross-validated accuracy. Stacking ensembles using a meta-learner (e.g., logistic regression or gradient boosting) trained on base classifier predictions could also yield substantial accuracy improvements.

8.2 Deep Learning Integration

Incorporating deep learning models — particularly transformer-based architectures adapted for tabular data (e.g., TabNet, SAINT) — may capture complex feature interactions that classical ML models cannot. Pre-trained biomedical language models could be fine-tuned if the symptom data is available in textual form.

8.3 Feature Engineering and Enrichment

Enhancing the feature set with demographic data (age, gender, geographic region), temporal symptom progression, and comorbidity information could dramatically improve prediction accuracy. Feature importance analysis using Random Forest or SHAP values can guide the selection of the most discriminative symptoms.

8.4 Addressing Class Imbalance

More aggressive resampling strategies such as SMOTE-ENN (combined over and undersampling) or class-weighted loss functions could improve minority class recall without significantly compromising majority class performance.

8.5 External Validation and Deployment

Future iterations of this system should be validated on independent clinical datasets to assess true generalizability. A web or mobile application interface could be developed for real-time symptom input and prediction, incorporating confidence scores and differential diagnosis lists.

9. Conclusion

This project successfully implemented and evaluated a comprehensive machine learning pipeline for disease prediction from symptomatic inputs across 38 disease classes. Four classification algorithms — Decision Tree, Random Forest, SVM, and Naive Bayes — were trained, cross-validated, and compared, alongside a combined ensemble model.

The Random Forest classifier emerged as the top performer with a test accuracy of 67.25%, followed by SVM and the Combined Model at 66.70%. Decision Tree achieved a cross-validation mean of 65.39%, while Naive Bayes significantly underperformed at 37.61% due to the violation of its conditional independence assumption in symptom-correlated medical data.

The analysis of confusion matrices provided valuable insights beyond aggregate accuracy metrics, revealing systematic misclassification patterns linked to class imbalance and overlapping symptom profiles. The combined ensemble model, while competitive, did not surpass Random Forest's standalone performance under simple majority voting, highlighting the need for more sophisticated ensemble strategies when constituent classifiers vary widely in predictive power.

This study contributes a structured comparative evaluation framework for multi-class disease prediction and identifies actionable pathways for future improvement. As AI-powered healthcare tools continue to mature, projects such as this lay the foundational groundwork for clinically deployable diagnostic support systems that could ultimately improve patient outcomes and healthcare access worldwide.

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